Resolution Geometry

Physical Curvature: The Canonical Connection as a Law of Motion

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Abstract

Resolution Geometry (RG) selects, for a regular partially observed system with a declared positive cost P, a unique horizontal lift

$$B = PJ^\top (JPJ^\top)^{-1},$$

the minimum-P^{-1} response to a requested observed velocity. The curvature of this canonical connection is valued in the null sector. Its holonomy displaces the unobserved configuration after any closed observed loop, by oriented area times curvature at second order.

This paper shows that this RG-native object is a *law of motion* for two classes of physical systems:

- *overdamped (Stokes) systems*, where P is the mobility;
- *free bodies with zero conserved momentum in the unobserved directions*, where P is the inverse inertia.

In both cases the response hypothesis H-resp holds exactly. RG then predicts physical motion from locally calibrated inputs alone:

1. an exact cross-coupled feedback apparatus with residual displacement $AB/\sqrt{L^2 + A^2}$, which is 9.9995 μm for $L = 100$ mm and $A = B = 1$ mm;
2. the three-sphere microswimmer, with curvature $(a/3)[L_1^{-2} + L_2^{-2} - (L_1 + L_2)^{-2}]$, whose prediction converges to an independent force-free simulation (ratio $1.046 \to 1.020 \to 1.010$);
3. a free three-link body reorienting with zero angular momentum (-0.275 rad for a 0.8×0.8 rad joint loop), where the RG lift equals the angular-momentum law to 2×10^{-10};
4. Purcell's scallop theorem, as flatness on reciprocal paths.

The decisive non-circular test uses the Einstein relation. Because B is invariant under $P \to cP$, the estimation connection fitted to *passive* Brownian fluctuations predicts *active* swimming that it never saw, to within 1.0–1.5% on five seeds.

No general relativity or quantum mechanics is imported. The physical inputs are a cost metric and a response rule, and both are locally testable. This is RG doing physics with its own geometry. It is not spacetime curvature, and it claims no priority over the gauge theory of deformable bodies.

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1 The standard

To have "business in physics", a framework must turn an object it defines into a falsifiable prediction about motion, from inputs calibrated *without* the predicted effect. Here the object is RG's canonical latent connection (print ladder L80), together with its curvature. The inputs are:

- the observation map F with Jacobian J;
- a positive cost P;
- the response rule H-resp.

Each can be calibrated by local, single-leg experiments. The predicted effect, the loop holonomy, is held out.

2 The connection and its curvature

Theorem 2.1 (Canonical lift; PC01)**.** *For J of full row rank and $P \succ 0$, the unique minimizer of $\frac{1}{2}w^\top P^{-1}w$ subject to $Jw = v$ is $w = Bv$, with $B = PJ^\top(JPJ^\top)^{-1}$ and $JB = I$.*

Proof. By Lagrange multipliers, $P^{-1}w = J^\top\lambda$ and $JPJ^\top\lambda = v$. Uniqueness follows from strict convexity. Numerically, the formula agrees with constrained optimization on 200 random instances. □

Definition 2.2 (RG transport curvature)**.** With $H_i = Be_i$, define $\Omega_{ij} = (I - BJ)[H_i, H_j]$. It is the Ehresmann (O'Neill) curvature of the canonical connection, valued in the null sector $\ker J$ [2, 3].

Theorem 2.3 (Second-order response law; PC02, PC10)**.** *For a closed observed rectangle of sides a, b traversed $1 \to 2 \to -1 \to -2$, the unobserved configuration is displaced by $ab\,\Omega_{12} + O(\varepsilon^3)$, while every observation returns exactly. Reversing the orientation reverses the sign.*

No-Go 2.4 (The observation map alone does not fix the curvature; PC03). For the same F, other lifts exist, and some of them commute and give zero holonomy. The curvature is locked by $(F, P,$ rule), not by observational equivalence. This is the ladder's no-law-natural-connection result (L79), now with a physical consequence.

Proposition 2.5 (Covariance; PC04)**.** *Under a nonlinear reparametrization of the latent configuration, transforming P as a tensor reproduces the same physical endpoint (difference 4×10^{-15}). Silently resetting P to the identity in the new chart changes it (by 1.2×10^{-3}). The holonomy belongs to (F, P), not to a chart.*

3 An exact apparatus

Take configuration (X, Y, Z) and feedback channels $U = X$ and $V = Y + XZ/L$, with an isotropic cost. Then

$$\Omega_{uv} = \frac{(0, -x, 1)}{(1 + x^2)^2}, \qquad J\Omega = 0.$$

For an observed rectangle of sides A, B, every observation returns exactly and

$$\Delta Z = \frac{AB}{\sqrt{L^2 + A^2}} = \frac{AB}{L} - \frac{A^3 B}{2L^3} + \cdots .$$

For $L = 100\,\mathrm{mm}$ and $A = B = 1\,\mathrm{mm}$, the prediction is $\Delta Z = 9.9995000375\,\mu\mathrm{m}$. Integration agrees to machine precision, the reversed loop flips the sign, the linear control gives zero, and the cubic coefficient $-\frac{1}{2}$ is resolved (PC10).

Protocol. The experiment proceeds in three steps:

1. calibrate J and P by single-leg responses, with no closed loop;
2. register ΔZ;
3. run the loop.

The prediction can fail in two ways, and each failure is informative:

- the apparatus does not implement H-resp, which single-leg tests detect;
- the calibrated P is wrong.

4 When the canonical connection is a law of motion

Hypothesis 4.1 (H-resp)**.** The system responds to a requested observed velocity with the minimum-P^{-1} lift.

Theorem 4.2 (Physical realization)**.** *H-resp holds exactly in two classes of systems:*

(a) Overdamped systems. *Generalized forces satisfy $\Gamma\dot{\xi} = f$, with no external force along $\ker J$, so $f \in \operatorname{im} J^\top$, and the observed coordinates are driven. Then $P = \Gamma^{-1}$ is the mobility, and the motion is the minimum-dissipation lift [5, 6].*

(b) Free bodies with zero conserved momentum. *The kinetic metric is M, and the momentum $K^\top M\dot{\xi}$ conjugate to the unobserved (symmetry) directions $K = \ker J$ is conserved and zero. Then $M\dot{\xi} \in (\ker J)^\perp = \operatorname{im} J^\top$, so $P = M^{-1}$, and the motion is the minimum-kinetic-energy lift: the mechanical connection [16, 15].*

Proof. In both cases $P^{-1}\dot{\xi} = J^\top\lambda$ and $J\dot{\xi} = \dot{q}$, which are exactly the Lagrange conditions of the canonical lift. □

Corollary 4.3 (Estimation equals physics; PC08)**.** *In thermal equilibrium, the Einstein relation gives diffusion covariance $2k_BT\,\Delta t$ times the mobility. The estimation connection (the covariance-weighted lift of RG's uncertainty layer, the noiseless Kalman limit) is therefore the physical connection, because B is invariant under $P \to cP$, and temperature drops out [7, 4].*

5 Three physical systems

5.1 The three-sphere microswimmer

Three beads of radius a sit on a line with arm lengths L_1, L_2, in Stokes flow with Oseen interactions. The swimmer is force-free, so H-resp(a) holds with P the mobility. The RG curvature, computed symbolically to first order in a, is purely translational:

$$\Omega = \frac{a}{3}\left[\frac{1}{L_1^2} + \frac{1}{L_2^2} - \frac{1}{(L_1+L_2)^2}\right].$$

This bracket has the functional form of the known leading-order swimming law [11, 12]. The per-cycle displacement is Ω times the enclosed shape area. It is compared with an *independent* force-free simulation, which solves for the bead forces directly (PC05):

a	d	simulated	RG prediction	ratio
0.05	0.1	9.584×10^{-4}	9.163×10^{-4}	1.046
0.025	0.05	1.169×10^{-4}	1.145×10^{-4}	1.020
0.0125	0.025	1.445×10^{-5}	1.432×10^{-5}	1.010

Every bead, and the centroid, is displaced equally (spread 2×10^{-12}; PC09), so the holonomy is a group element.

Corollary 5.1 (Scallop theorem as flatness; PC06)**.** *A reciprocal stroke retraces its path and encloses no area, so the holonomy vanishes at every amplitude. In-phase strokes and the two-sphere swimmer both give* $|\Delta| < 10^{-15}$ *[8]. Without hydrodynamic interactions the connection is flat and nothing swims at all: the curvature* is *the interaction.*

5.2 A falling-cat body

A free planar three-link chain (four unit point masses, links of 0.3 m) carries zero angular momentum. H-resp(b) holds with P the inverse inertia. The RG lift equals the angular-momentum law at every sampled state (difference 2×10^{-10}). A 0.8×0.8 rad square loop of the two joint angles, based at $(1.5, 1.5)$, reorients the body by -0.2748 rad, and the same square traversed backwards gives $+0.2748$ rad. The rotation per area converges to the curvature (-0.228, then -0.222, as the loop shrinks). A lift with P the identity predicts no rotation at all, and is falsified (PC07). This is the gauge-theoretic falling cat [14, 15], obtained as RG holonomy.

5.3 Fluctuations predict locomotion

The connection is calibrated from *passive* Brownian increments only: at 81 shapes, the latent increments are regressed on the observed ones, with no driving and no loop. The resulting connection predicts the *active* swimmer's per-cycle displacement. The predictions are 2.840–2.927×10^{-3}, against a simulated 2.883×10^{-3}, an error of 1.0–1.5% over five seeds (PC08). Calibrating without hydrodynamic interactions predicts no swimming and is caught by the canary. This is RG's estimation layer and its physics layer as one object.

6 What is RG's, and what is not

RG's.

- One object, $B = PJ^\top(JPJ^\top)^{-1}$, is the canonical latent connection of the observation quotient (L80). It is simultaneously the noiseless Kalman lift, the minimum-dissipation motion of Stokes bodies, and the mechanical connection of free bodies.

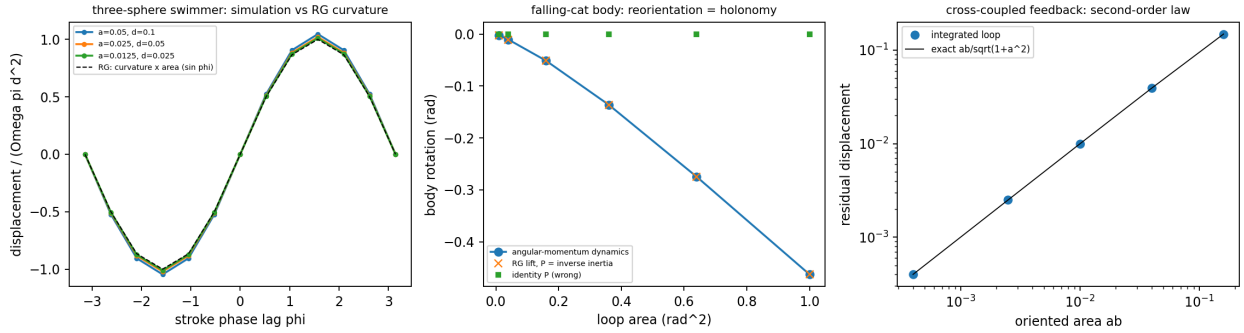


Figure 1: Left: three-sphere swimmer. The simulated displacement, normalized by curvature times area, against the stroke phase lag, converges to $\sin\varphi$ as the bead radius and amplitude shrink. Centre: falling-cat body. The rotation against loop area agrees for angular-momentum dynamics and the RG lift; the identity-P lift fails. Right: the cross-coupled apparatus follows the exact second-order law.

- The Einstein relation makes the estimation and physics instances coincide, so geometry calibrated from fluctuations predicts locomotion.
- The typing: LOCKED (conditional) given $(F, P, \mathsf{H\text{-}resp})$; CORRESPONDENCE without P (L79); flatness on reciprocal paths.
- The calibrate-locally, predict-the-loop protocol.

Not RG's. The swimmer and falling-cat physics, and their gauge-theoretic description, are established [9, 10, 15, 16]. Overlap with geometric-phase theory is expected [17]. No priority is claimed.

Non-claims.

- This is not spacetime curvature.
- The law is second order in amplitude; it is not a force law.
- The physical inputs (a cost metric and a response rule) are imported from classical mechanics, hydrodynamics and statistical mechanics. They are not imported from GR or QM, and they are locally testable.
- The results are simulations within declared models (Oseen hydrodynamics, point masses), not experiments.
- The swimmer coefficient is derived and validated here against simulation. The literature is cited for the functional form.

7 Experimental programme

1. *The cross-coupled feedback apparatus.* Calibrate J and P by single legs, register ΔZ, then run the loop.
2. *Colloidal three-bead swimmers.* Optically driven swimmers have been realized [13]. Calibrate from passive fluctuations (PC08), then predict the driven displacement.
3. *Free-floating robots and satellites.* Predict reorientation from inertia calibrated by single-joint motions.

Each is a held-out prediction from RG geometry with locally calibrated inputs. Each failure locates either H-resp or P.

A Reproducibility

`rg_physical_curvature_attack.py` (PC01–PC11) passes 11/11 on seeds 20261027, 1, 42, 777 and 2026. Its canaries are a curvature claimed for the commuting lift, and calibration without hydrodynamic interactions; both are caught. Corrections before registration:

1. The falling-cat "reversed loop" first used a negative side length, which traverses a different square with the same orientation. It is now the same square traversed backwards.

2. The reference-point gate was first tighter than the integrator tolerance.

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